Jeff Shun Qu

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Education

University of Toronto September 2023 - May 2028 (Expected)

Bachelor of Applied Science, Computer Engineering + PEY Co-op

Relevant Coursework: Algorithms & Data Structures, Operating Systems, Applied Fundamentals of Deep Learning, Introduction to Artificial Intelligence, Linear Algebra

Skills

Languages: Python, C/C++, JavaScript, SQL

ML/Data: PyTorch, NumPy, Pandas, Feature Engineering, Model Evaluation, Deep Learning

Systems & Robotics: Linux (Ubuntu), ROS2, Raspberry Pi, Git, Sensor Integration, Debugging

Software Development: FastAPI, REST APIs, MySQL, Apache Solr, JSON, Unit Testing, Agile Collaboration

Experience

Research Assistant | Python, Javascript, FastAPI May 2024 - August 2024

University at Buffalo Buffalo, New York

- Developed Python data-processing pipelines to transform large-scale linguistic datasets into structured JSON for Apache Solr indexing and downstream search applications.
- Designed RESTful APIs with FastAPI to expose indexed data through low-latency backend services.
- Built JavaScript frontend components integrated with backend APIs to support interactive phoneme search and real-time data exploration.
- Collaborated with a multidisciplinary research team to develop and debug an end-to-end software platform integrating data pipelines, backend services, and user interfaces.

Projects and Leadership

Design Team – Subsection Lead | ROS2, Python, Linux, Raspberry Pi, Git September 2025 - January 2026

University of Toronto Underwater Exploration Design Team Toronto, Ontario

- Led a team of three students developing modular Python drivers and real-time sensor pipelines for camera, IMU, and magnetometer systems on an autonomous underwater robot.
- Designed reusable software interfaces that simplified integration between hardware sensors and robotics software components.
- Optimized camera streaming bandwidth from **15 MB/s** to approximately **2 MB/s** while maintaining **30 FPS**, improving performance for real-time robotic perception.
- Debugged Linux software and hardware integration issues to improve reliability across multiple sensors.

Machine Learning xFIP Prediction Model | Python, PyTorch, NumPy, Pandas July 2025

- Developed a two-stage deep learning pipeline using PyTorch to model high-dimensional baseball pitch-tracking data and predict season-level expected xFIP.
- Engineered predictive features and iteratively refined neural network architectures through quantitative evaluation, reducing prediction error by **17%** over the baseline.
- Built reproducible training, validation, and experimentation pipelines supporting rapid model iteration and performance analysis.

GIS Navigation Project | C++, GTK/Glade, Git April 2025

- Developed an interactive GIS application using OpenStreetMap data and C++ to compute efficient navigation routes.
- Implemented parallel Ant Colony Optimization, improving routing performance across large graph datasets.
- Designed an interactive GTK interface enabling visualization and debugging of computed routes.